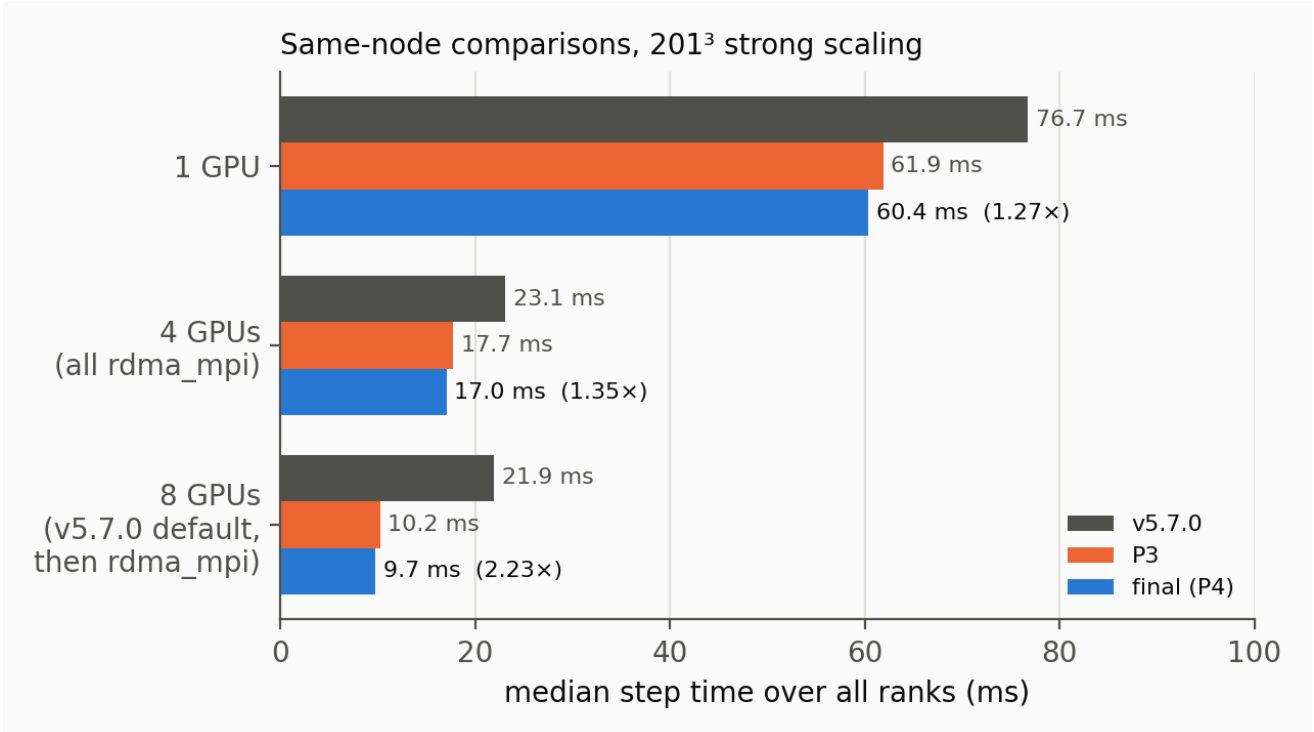
Making MFC faster on H200 without changing its answer

CS5434 High Performance Computing, Coding-Adv-HW-1

MFC v5.7.0, SC26 Student Cluster Competition application, nano4 (8× H200)

Result: 1.27× to 2.23× faster, bitwise identical



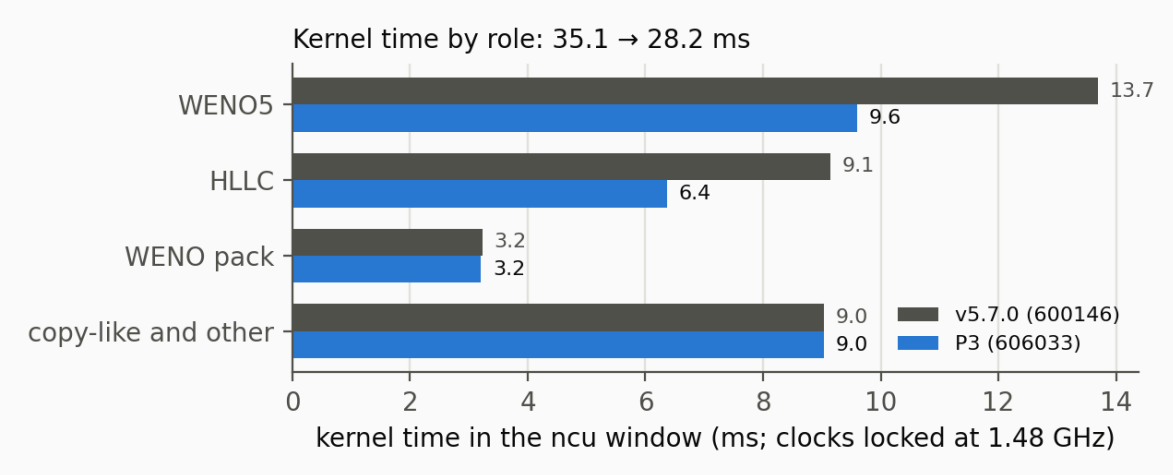
201³ cells, FP64, 30 steps. Each comparison on one node. Jobs 606017 / 621788, 606018 / 622305, 606019 / 622306.

Six changes, no physics changed

Change	Problem	Fix	Gain
P1	host sync per boundary call	async, wait once	−1.4 %
K4b	Riemann helpers not inlined	one translation unit	−7.3 %
K2	WENO at 196 registers	`maxregcount:128`	−10.6 %
P3	~1.2 ms GPU idle per stage	async kernels	−1.7 %
P4	WENO on unread ghost cells	interior only	−2.5 %
`rdma_mpi`	halos through host	GPU→GPU over NVLink	−27 % (4 GPUs)

Step time on 1 GPU, 77.06 → 60.38 ms. P3 and P4 gain more at 8 GPUs (P4: −4.0 %).

Why it is faster



ncu 600146 (v5.7.0) vs 606033 (P3)

- WENO5: occupancy doubled. HLLC: helpers inlined.
- Still slow: copy kernels at a third of HBM bandwidth, halo not overlapped.

Correctness and limits

- Every run: final state bitwise identical to v5.7.0.
- `./mfc.sh test`: 690 pass, 23 fail, 0 new. The same 23 fail on unmodified v5.7.0.
- Tried and rejected: `vector_length`, register cap 96, NVHPC 25.9.
- Next: overlap halo with compute, fuse copy kernels, send PRs upstream.

Report `report.md`, patches `patches/opt/`, raw data `results/`